

WEEK 1 • SESSION 1

Enterprise Data Engineering, Analytics & AI

Role of Data in Enterprise Systems

Why data is the backbone of every modern organisation



Learning Objectives

Understand what data IS and why enterprises depend on it

Identify the key roles data plays across different business functions

Recognise how data moves through an organisation (data lifecycle)

Distinguish structured, semi-structured and unstructured data

Explore real-world examples from retail, banking, healthcare & logistics

What is DATA?



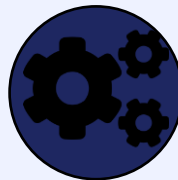
Raw Facts

Numbers, text, images, events —
anything that can be recorded



Insights

Data processed into meaning — tells
you WHAT happened & WHY



Decisions

Insights drive strategy — product
launches, pricing, hiring

Data is Everywhere in Business

Every department creates, consumes and depends on data every single day



HR

Headcount, attendance, payroll



Finance

Revenue, costs, forecasts



Operations

Inventory, production, logistics



IT

Logs, tickets, uptime metrics



Sales

Pipeline, conversions, quotas



Marketing

Campaigns, CTR, customer segments

Three Types of Enterprise Data

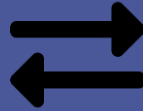


Structured

Organised in rows & columns

- SQL Tables
- Excel sheets
- ERP exports

"Like a perfectly organised filing cabinet"



Semi-Structured

Has some organisation but no fixed schema

- JSON
- XML
- Email headers

"Like folders that contain documents of varying formats"



Unstructured

No predefined model or format

- Videos
- Images
- Social media posts

"Like a warehouse of mixed goods with no labels"



QUICK QUIZ TIME!

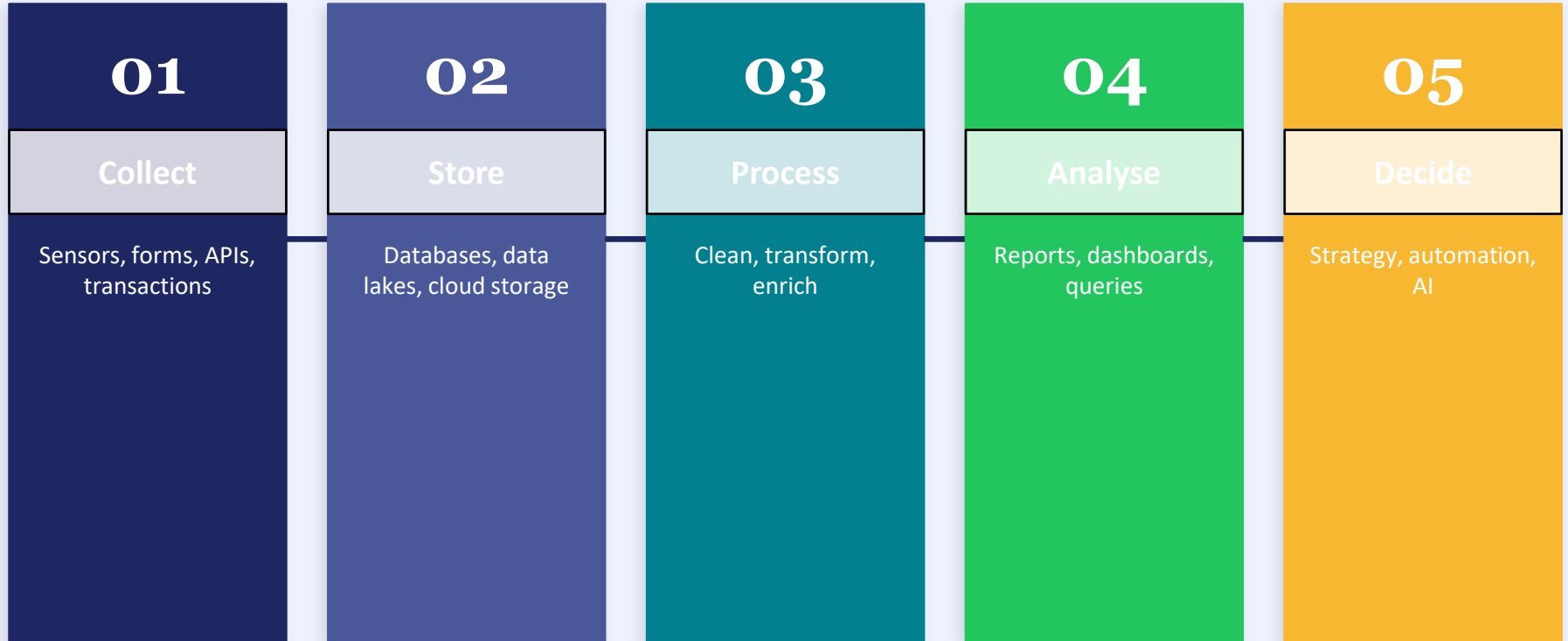
Q1 → What are the three types of data?

Q2 → What is the difference between data and an insight?

(Answers on next slide — don't skip ahead!)

The Enterprise Data Lifecycle

How data travels from birth to business value



Real-World: Data in Action

Retail

Amazon uses purchase history + browsing data to recommend products

Data: Clickstream, orders, inventory

✓ **35% of revenue from recommendations**

Healthcare

Hospitals track patient vitals in real-time to predict deterioration

Data: IoT sensor data, EHR records

✓ **Early interventions, fewer ICU admissions**

Banking

Banks flag unusual transactions as potential fraud within milliseconds

Data: Transaction logs, location, behaviour

✓ **Millions saved in fraud prevention daily**

Logistics

Delivery companies use GPS + traffic data to optimise routes

Data: GPS, traffic APIs, package data

✓ **Faster deliveries, lower fuel costs**

Why Data Quality Matters

Bad data → Bad decisions → Real cost



\$12.9M

Average annual cost of bad data
per organisation (IBM)



80%

of a data analyst's time spent
cleaning dirty data



1:10

Cost ratio: preventing a data error
vs fixing it later

● Activity: 'Can you think of a decision your company made that may have been affected by wrong data?'

The Data Professional Landscape

Roles that exist because of enterprise data needs

Data Scientist

ML models, predictions, statistics

Adjacent skill

Data Analyst

Queries data, creates reports

Related role

Data Architect

Designs data systems & governance

Advanced path

Data Steward

Quality, governance, compliance

Awareness
needed

Session Summary



Data is raw facts — processed data becomes insights that drive decisions



Every enterprise department creates and consumes data daily



Data exists as structured, semi-structured, and unstructured formats



The data lifecycle: Collect → Store → Process → Analyse → Decide



Data quality is critical — bad data costs organisations millions annually



Data professionals (you!) are in high demand across all industries

» Next: S2 — Data Flow in Applications (Input → Storage → Output)

Questions?

No such thing as a bad question — only missed opportunities to learn!



Before next session:

Observe 3 types of data used in your daily life
Note which department in a company might use them